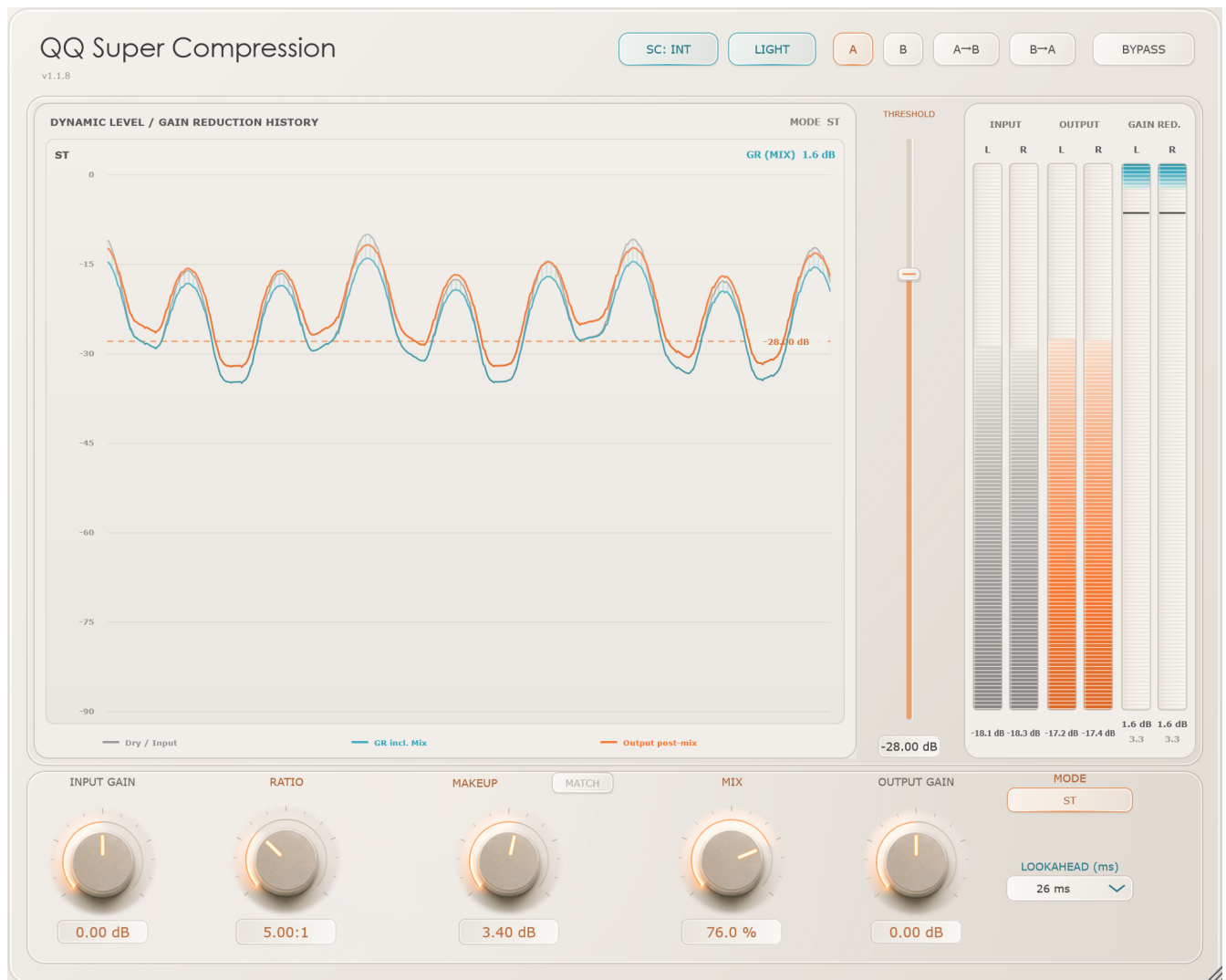


QQ Super Compression

USER MANUAL / 1.1.8

Control dynamics. Keep the source's attack.

For vocals, instruments, buses and mastering. Use Lookahead, Ratio, Threshold and Mix to shape dynamics without conventional Attack / Release controls.



Start with what you hear

Decide how steady the dynamics should be and how much articulation to retain. The Display helps explain changes; it does not replace listening.

Quick start and navigation

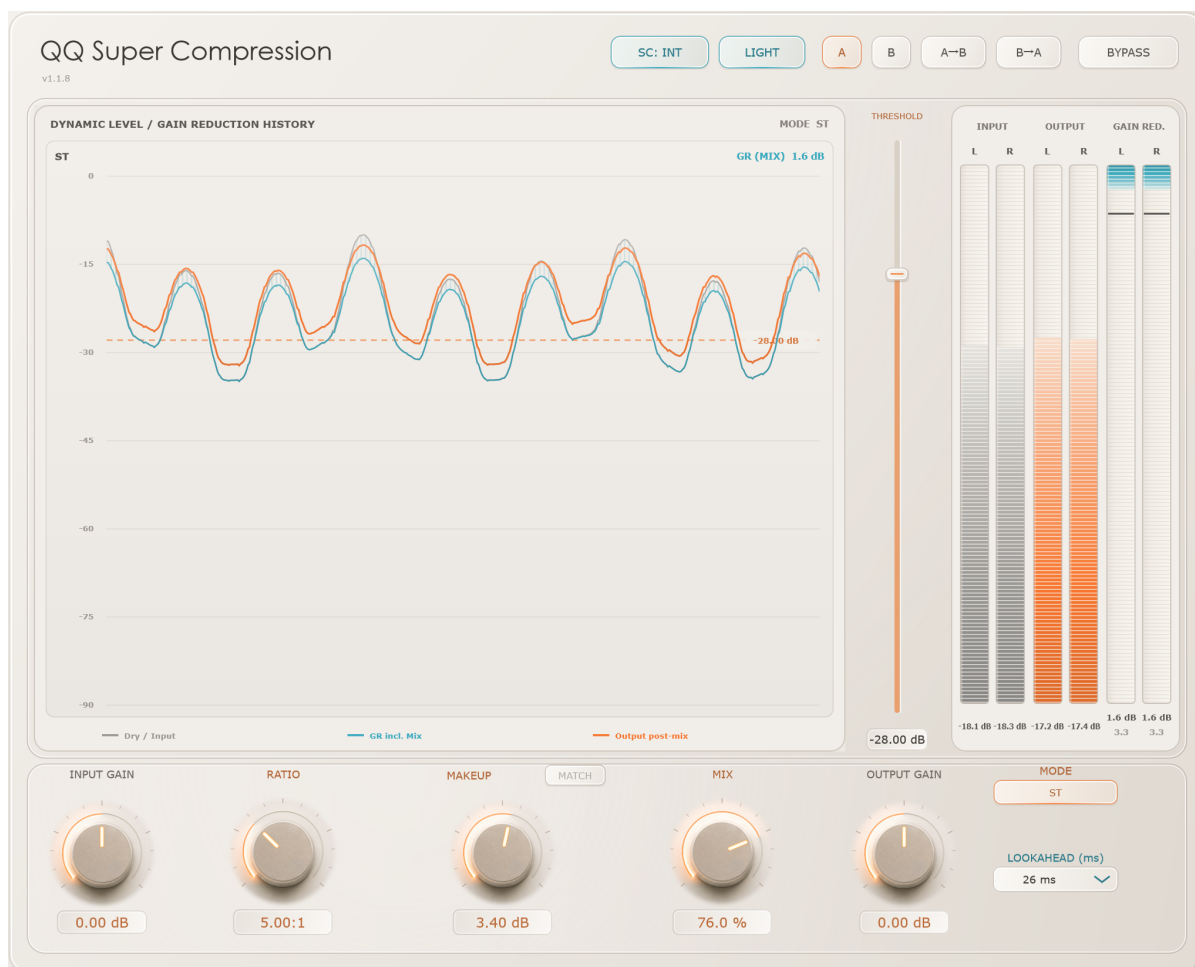
A first session in six steps.

- 1 **Insert on the track to process**
Start with ST and SC: INT so the plug-in responds to that track's own signal.
- 2 **Choose Lookahead**
26 ms is a practical starting point, not a rule for every source.
- 3 **Set the compression amount**
Adjust Ratio, then Threshold if you want an operating floor. Leave Threshold OFF for threshold-free processing.
- 4 **Use Mix to control compression**
If processing is too strong, lower Mix to blend back the original dynamics. GR changes with Mix too.
- 5 **Compensate level and compare**
Play a representative passage, use Match, then trim Makeup / Output Gain to avoid loudness bias.
- 6 **Listen in the full mix**
Use Bypass or A/B to judge articulation, consistency and groove - not only the GR number.

Find a topic	Pages
Interface, themes and gestures	3-4
Compression, level controls, Display, Lookahead	5-9
LR / MS and LINK	10-12
Sidechain panel, internal HPF, external routing	13-15
Mixing examples, parameter reference, help	16-18

Know the interface

The same audio controls remain in place across all themes.



Light / ST. The values illustrate the interface; they are not a recommended preset.

Area	Use it to
Display	Compare original dynamics, Mix-aware GR and final output.
Threshold	Choose the detector level where compression begins.
Input / Output / GR	Read input/output levels and effective compression.
Lower controls	Adjust Input, Ratio, Makeup, Mix and Output.
Mode / LINK / Monitor	Choose channel processing, link controls or audition domains.
Top right	Open sidechain, change theme, use A/B and Bypass.

Themes and everyday controls

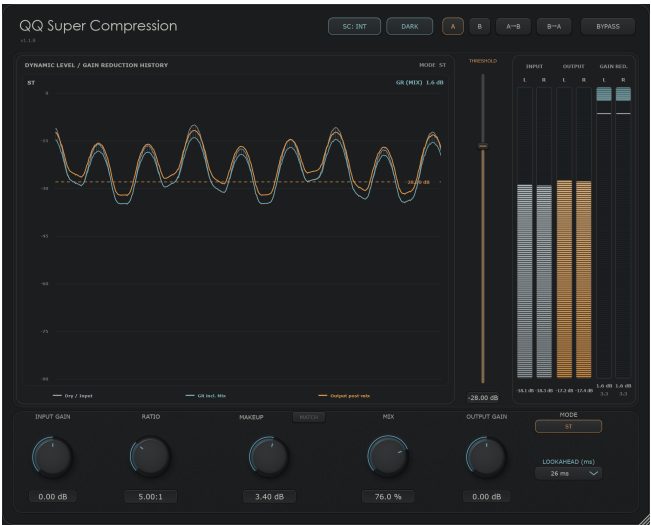
Click the top-right button labelled LIGHT, DARK or CLASSIC to cycle themes. The last selection is restored when the plug-in opens again.



The theme button is between SC and A/B.

Theme	Appearance
Light	Light interface.
Dark	Subdued dark interface.
Classic	Original classic dark interface.

All themes share the same layout, parameters and sound. Choose the appearance that suits your working environment.



Dark

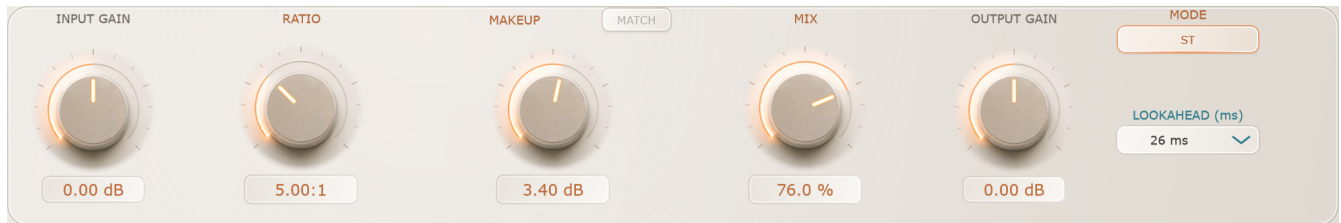


Classic

Gesture	Result
Drag / Shift-drag	Normal / fine adjustment.
Double-click a value	Type an exact value.
Alt + left click	Reset that knob to its default.
Ctrl/Cmd + Z	Undo; add Shift to redo.

Ratio, Input and Mix

Three useful controls, not interchangeable settings.



Ratio

Ratio sets compression strength. With the source and other settings held constant, increasing it generally produces more GR. Start here to ask for more or less compression. Its scale is not the fixed above-threshold dB slope of a conventional compressor, so do not transfer ratio numbers blindly.

Input Gain

Input Gain changes the level entering the processing path, including the Dry path. With INT it also changes the detector level relative to Threshold. With EXT, the external send and Key Gain set the detector level; Input Gain is not a substitute.

Mix

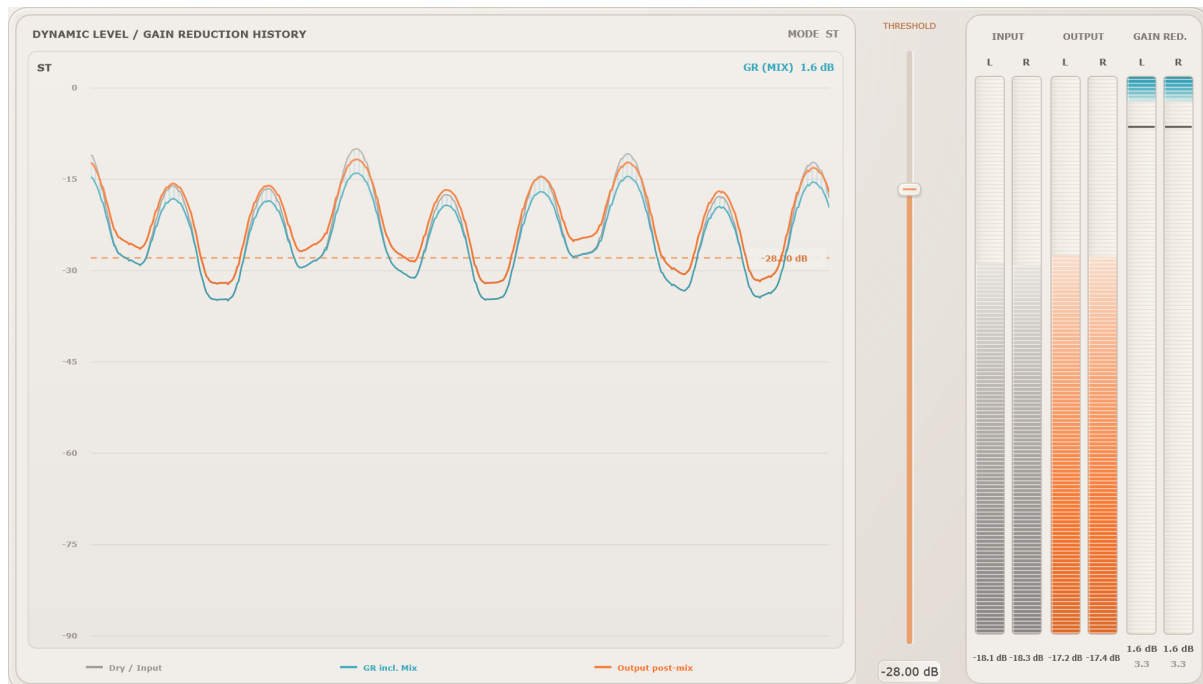
Mix blends Dry with processed Wet. It is an important compression control, not merely an effect amount: establish a clear compression character, then lower Mix to bring back as much original dynamics as needed.

GR includes Mix

At 0% Mix, effective GR is 0 dB; at 100% it reflects the full compression. Intermediate values are not simply the GR number multiplied by Mix. Makeup and Output Gain are excluded from GR.

A practical order: find the compression character with Ratio, set its final strength with Mix, adjust Input Gain when you need a different input level, then compare loudness again.

Threshold: choose the operating floor



The dashed line is Threshold; use the adjacent slider or value box to adjust it.

Setting	What it means in use
OFF	Use threshold-free processing. Signal level, Ratio and Mix still affect the amount.
Threshold enabled	Compression begins above the selected detector level; lowering the threshold generally includes more of the signal.

- 1 Play the passage that needs control**
Include stronger events as well as quieter detail.
- 2 Find the useful range**
Move from a higher threshold downward, notice where GR appears, then use Ratio and Mix for strength.

This is not Attack / Release

Threshold chooses which detector levels are compressed; it does not set build-up or recovery time. With EXT it follows the external key, not the current track's Dry trace.

Match loudness before comparing

Control	Use
Makeup	Compensates Wet level before Mix. At lower Mix it is not a global output trim.
Output Gain	Trims the final output without changing the GR reading.
Match	Makes a one-time loudness adjustment to Makeup from played material; it is not continuous auto gain.

- 1 Play representative material**
Wait until Match is available, then click it. Very short or silent passages may not provide a usable reading.
- 2 Check final output loudness**
Match compares Dry with Wet before makeup; the final mixed output may still need a small Makeup or Output Gain correction.
- 3 Compare sound, not level**
Toggle Bypass and listen to articulation, tails and consistency across the passage.

Use A / B for two alternatives

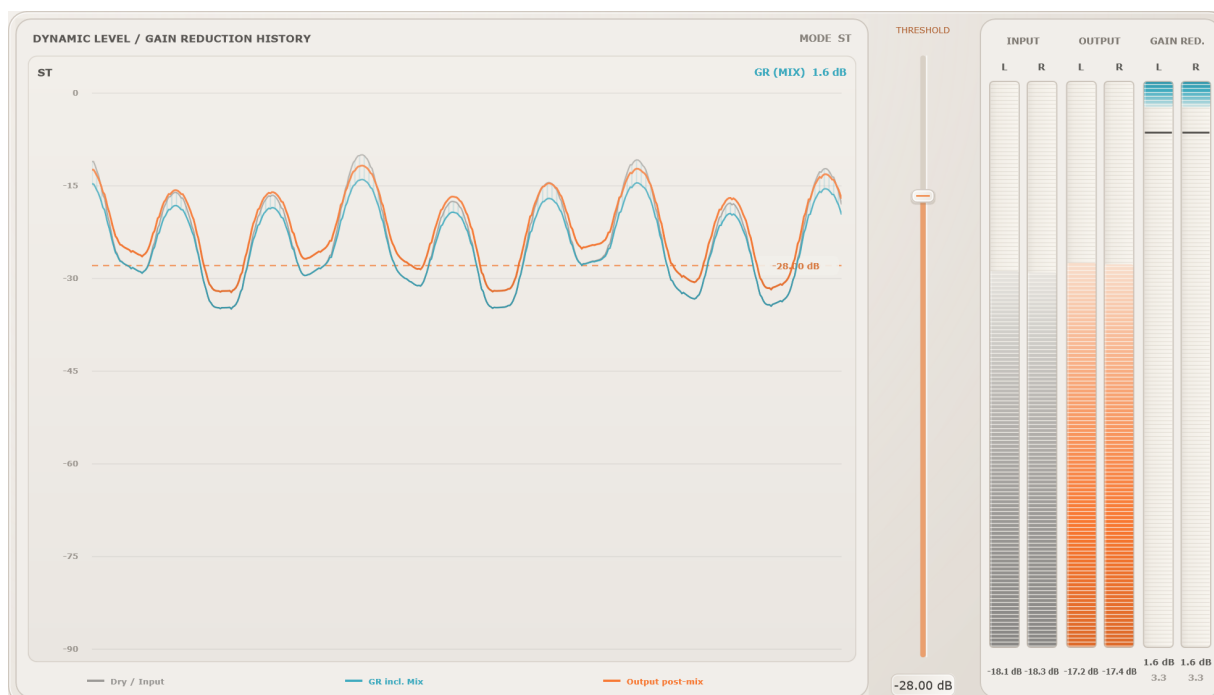
Set up A, copy with A→B, then switch to B and try another approach. Click A or B to compare. B→A copies in the other direction; copying replaces the destination's sound settings.

A/B includes sidechain Source, Key Gain and HPF. Theme is not a sound setting; Bypass, LINK and SC LISTEN are not recalled by the A/B sound snapshot.

0% Mix is not full Bypass

At 0% Mix, Input Gain and Output Gain still apply. Bypass compares the original signal without those gain and compression stages.

Read the Display and meters



What you see

Meaning

Dry / Input

Original dynamics before Input Gain; the grey line does not move as a whole when Input changes.

GR incl. Mix

The effective reduction boundary. Its distance below Dry represents Mix-aware attenuation, excluding Makeup / Output Gain.

Output post-mix

Final output including Makeup, Mix and Output Gain.

External key

With EXT and an available external bus, a faint cyan trace shows the sidechain reference. Compare it with GR to check when the key triggers compression.

Gain Red. Meter

Zero is at the top; downward growth means more reduction. A short hold helps reveal recent stronger reduction.

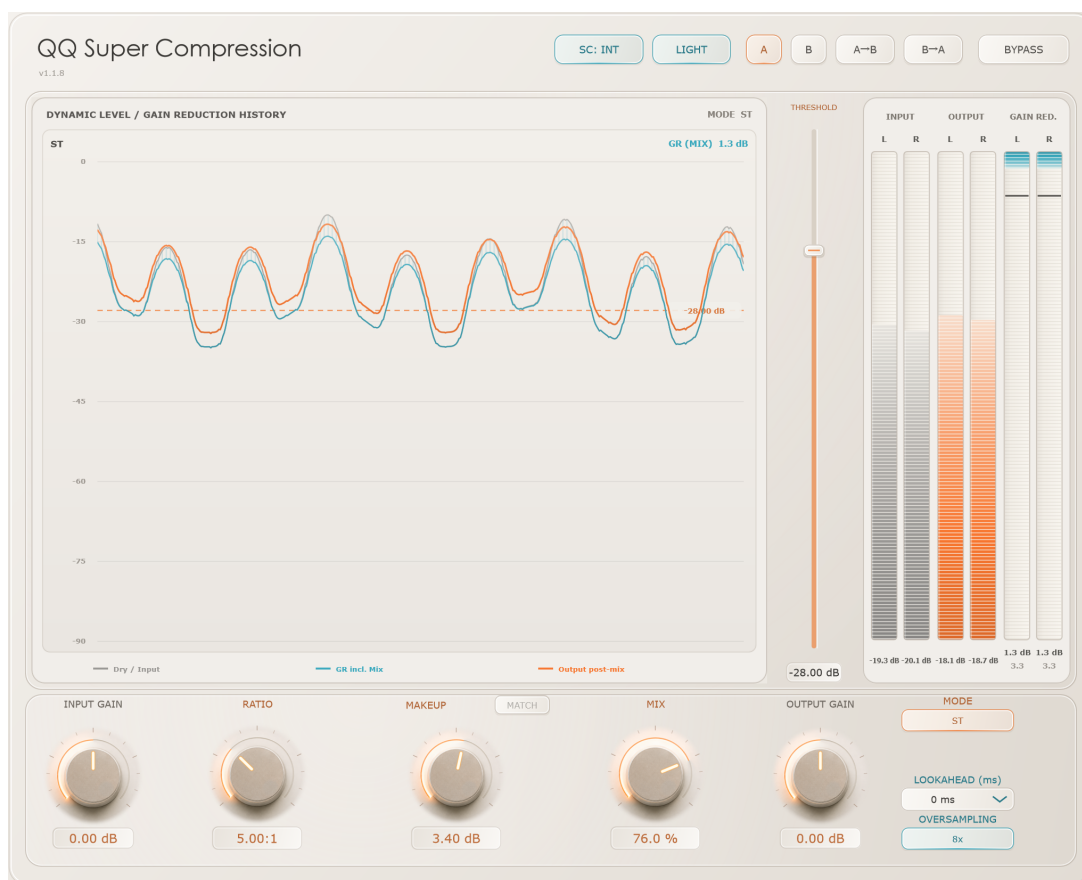
Light / Dark use grey Input, orange Output and cyan-blue GR. Classic uses yellow Output and pink GR. The meanings are the same.

History responds to current settings

Input, Ratio, Threshold, Mix, Makeup and Output Gain update the visible history. EXT Key Gain updates live too; HPF refreshes after release and may show HPF UPDATING. The graph is a history reference under current settings, not an audio replay.

Lookahead and the 0 ms option

Lookahead is an advance observation window, not Attack. Longer settings generally suit cleaner, steadier control but add plug-in latency, which the host normally compensates.



At 0 ms, Oversampling appears below Lookahead.

Choice	Use and trade-off
10 ms	Shorter window and lower latency.
26 / 40 ms	Useful starting points; choose by low-frequency stability and the overall sound.
80 / 100 ms	Longer windows; listen for greater stability and consider the extra latency.
0 ms	More pronounced nonlinear colour, not the most transparent mode.

At 0 ms choose 1x / 8x / 16x Oversampling. Start with 8x; higher rates cost processing and latency. The control is hidden at 10 ms and above. 0 ms with 8x / 16x still adds latency.

ST and LR: together or independently

Click Mode at the lower right to cycle ST → MS → LR. Start with ST; choose LR when the two sides genuinely need different treatment.



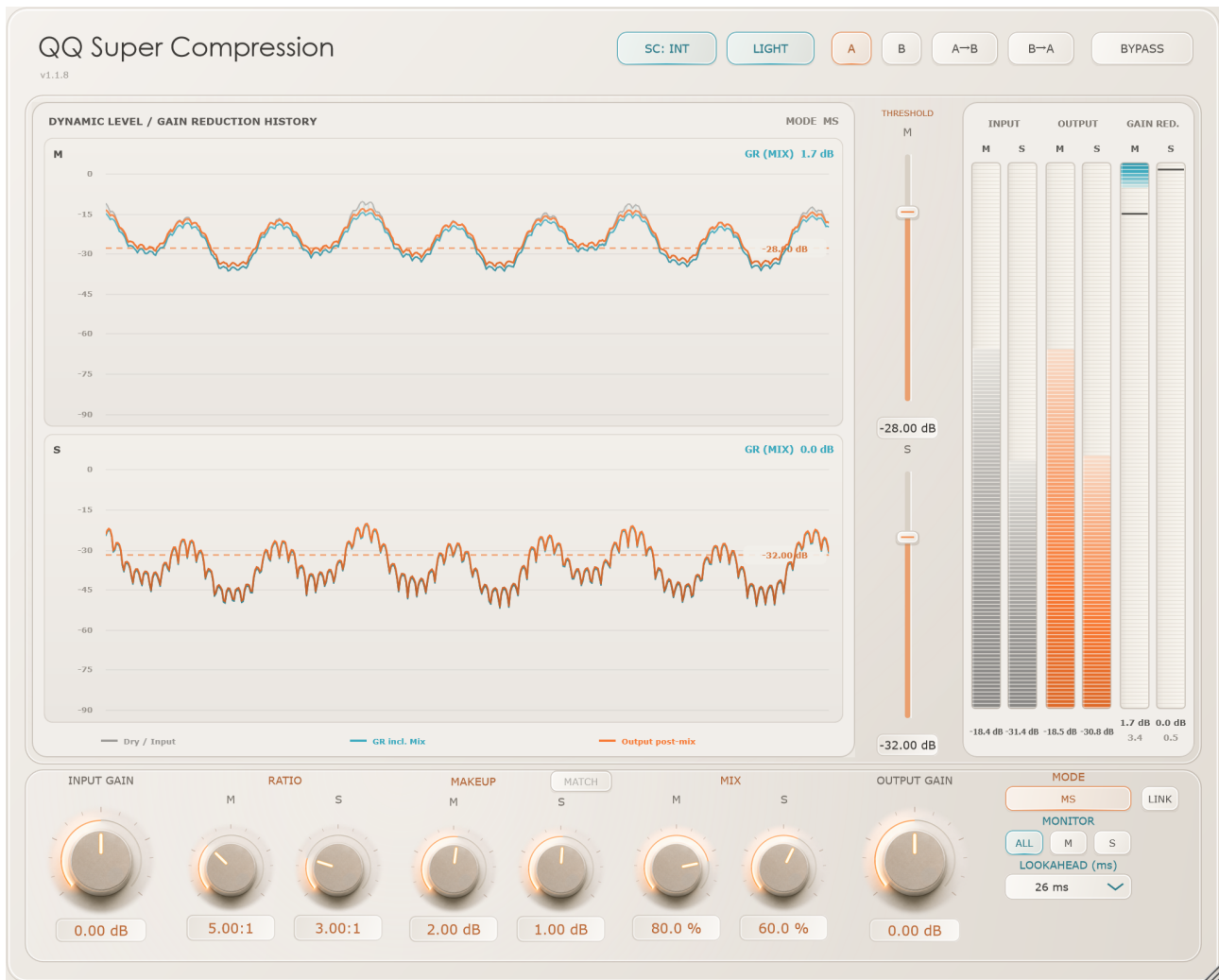
LR: L above, R below, with independent controls.

Mode	How to use it
ST	Uses common compression gain for stereo; a useful starting point for tracks and buses.
LR	Set Ratio, Threshold, Makeup and Mix independently for Left/Right. On asymmetric material, also check image balance.

MONITOR

Choose L or R to audition that side centred. Return to ALL to check normal stereo, and do not leave one-sided audition active for the final output.

MS: control centre and sides



MS: Mid above, Side below.

Mid mainly represents content common to both sides; Side represents their differences. They are not a precise separation of a centre track from every other track.

- 1 **Listen to M and S first**
Use M / S under MONITOR, then return to ALL.

- 2 **Adjust each for a reason**
For steadier centre dynamics, start with M Ratio / Mix. Treat S cautiously when preserving ambience and width.

Recheck the full mix

Compression or level changes in M/S can alter the sense of space. In ALL, check centre, width and mono compatibility, not only soloed domains.

LINK: adjust together, keep the difference

In LR or MS, click LINK next to Mode. Set each domain first, then enable LINK to adjust both without forcing them to identical values.



Example: LINK is on while the two Ratio values remain 3:1 and 5:1.

Pair	Before	After an equal change
Ratio	3:1 / 5:1	4:1 / 6:1
Threshold	-20 / -10 dB	-18 / -8 dB
Makeup	-3 / +1 dB	-2 / +2 dB
Mix	100% / 70%	90% / 60%

1 Set the relationship with LINK off

Decide which domain needs more compression or a different blend.

2 Enable LINK for joint adjustments

Move either parameter in a pair; the other follows by the same change. Fine adjustment and direct entry work too.

Both stop at a limit

If either side would exceed its range, both stop where the difference can be retained. One Threshold at OFF cannot form a finite dB offset; enable both thresholds first if you want that relationship.

Sidechain: start with the panel

Click SC: INT / SC: EXT at the top right. Its label shows the current source.



Control	How to use it
INT / EXT	INT responds to the track itself. EXT uses another track's signal to control this track.
KEY GAIN	Available only with EXT. Adjusts the external detector level and trigger strength, not the main track level.
HPF	Reduces low frequencies in the INT or EXT detector key. OFF is full-band; active range is 20-500 Hz.
KEY LEVEL	Check that the detector receives a signal. With EXT showing N/A, first check that the host sidechain bus is enabled.
SC LISTEN	Temporarily audition the detector key to check routing and HPF. Switch it off before judging normal output.

Two common tasks

The whole mix ducks whenever bass arrives: use INT + HPF (page 14).

The bass should make room when the kick hits: use EXT (page 15).

Internal sidechain: reduce bass dominance

Use this when kick or bass repeatedly pulls down the rest of a drum bus, mix bus or master. The goal is less low-frequency triggering, not less bass in the audible signal.

- 1 Start with INT and hear the problem**
Loop a passage with bass and other content together. Leave HPF OFF to establish a comparison.
- 2 Open the sidechain panel**
Click SC: INT and keep SOURCE on INT. A disabled Key Gain is normal here.
- 3 Raise HPF gradually**
Start at a low cutoff and listen for less whole-mix ducking on bass events. Make small changes.
- 4 Audition the key if needed**
Use SC LISTEN to hear which bass content HPF removes, then turn it off and return to the full sound.
- 5 Inspect history after release**
HPF history refreshes after you release the knob. If HPF UPDATING appears, wait for it to clear before comparing.
- 6 Rebalance the compression amount**
Filtering may reduce GR or change its shape. Revisit Ratio, Threshold or Mix as needed, then compare at similar loudness.

It is normal not to hear bass being cut

HPF filters only the key used to decide compression, not normal output. It is not equivalent to Key Gain: Key Gain scales the detector level; HPF changes frequency weighting. Too high a cutoff can make the compressor ignore bass that still needs control.

External sidechain: kick-controlled bass

Insert the plug-in on the bass - the track to turn down. Send the kick to its sidechain as the trigger. Do not reverse the tracks or confuse a normal audio send with a sidechain send.



- 1 Enable the host sidechain input**
 Enable sidechain input for this plug-in instance on the bass. Add and enable a kick send to that sidechain. Names and locations vary by host.
- 2 Select EXT in the plug-in**
 Open SC, choose EXT and play the kick. Check KEY LEVEL first; if it has no reading, fix routing before increasing Ratio.
- 3 Confirm the actual trigger**
 Briefly enable SC LISTEN: you should hear the kick, not the bass. Turn audition off afterward.
- 4 Set trigger level and reduction**
 Leave HPF OFF initially. Set detector strength with Key Gain or send level; use Ratio and Threshold for amount and operating range.
- 5 Use Mix to decide how much room to make**
 Find an audible duck, then reduce Mix to retain the desired bass sustain. Listen to both tracks together, not only solo bass.
- 6 Decide whether HPF helps**
 Raising HPF reduces low-frequency kick triggering and may weaken the duck you want. Use it when you want more emphasis on the attack or less unwanted low-bass triggering.

There are no conventional Attack / Release controls. Ducking shape depends on the trigger, Lookahead and compression settings; it is not guaranteed to be smoother than every conventional sidechain.

Apply it in a real mix

Vocals, guitar, piano: consistency with articulation

Try ST / INT and 26 ms as a starting point. Set Ratio, then use Mix to keep consonants, picking or hammer attacks natural. Enable Threshold when only stronger parts need control. After Match, compare in context: neither small nor large GR is a goal by itself.

Mix bus / mastering: a more cohesive balance

When you want a steadier, more cohesive mix without deliberately reshaping drum attacks through conventional Attack / Release, try it as an alternative to G Bus-style compression. Start gently, compare balance across sections and lower Mix when needed.

If every kick pulls the whole mix down, try INT + HPF. Do not raise the cutoff merely to make the GR number smaller. At similar loudness, check drum attack, low end, space and the song's larger dynamic movement.

Vocal-keyed accompaniment: make space for words

Insert on the accompaniment bus, send the vocal to its external sidechain and select EXT. Confirm the source with SC LISTEN; set Key Gain, Threshold and Ratio so the accompaniment steps back during vocals, then use Mix to control the depth.

If plosives trigger too much ducking, try raising sidechain HPF. If breaths or noise also make the accompaniment dip, check the sent signal and Threshold; HPF alone will not solve every unwanted trigger.

For every comparison

Turn SC LISTEN off, return MONITOR to ALL, and compare at similar loudness. Use an appropriate limiter when you need a strict final peak ceiling; this plug-in is not a brick-wall limiter.

Parameter reference

Parameter	Range / choices	Notes
Input Gain	-24 to +24 dB	Level into the processing path.
Ratio	1:1 to 32:1	Default 8:1; not a conventional dB slope.
Threshold	OFF / -119.99 to 0 dB	Default OFF; independent ST, L/R, M/S values.
Makeup	-36 to +36 dB	Default 0 dB; Wet level before Mix.
Mix	0 to 100%	Default 100%; included in effective GR.
Output Gain	-24 to +24 dB	Default 0 dB; final output level.
Mode	ST / MS / LR	Click to cycle.
LINK	OFF / ON	Relative control linking in LR / MS.
MONITOR	ALL / L / R or M / S	LR / MS audition; individual domain centred.
Lookahead	0 / 10 / 26 / 40 / 80 / 100 ms	Initial starting point 26 ms; new instances remember the last choice.
Oversampling	1x / 8x / 16x	Visible only at 0 ms; initially 8x.
SOURCE	INT / EXT	Default INT; EXT needs host routing.
KEY GAIN	-24 to +24 dB	Default 0 dB; adjustable only with EXT.
HPF	OFF / 20 to 500 Hz	Default OFF; filters only the detector key.
SC LISTEN	OFF / ON	Temporary audition; closing panel/editor switches it off.

Troubleshooting

Symptom	First check
EXT has no effect	Is the host sidechain input/send enabled? Is SOURCE on EXT? Check KEY LEVEL: N/A means no external bus is available; -inf commonly means silence or no incoming signal.
Key Gain is disabled	This is normal with INT. Key Gain adjusts EXT only.
HPF makes little audible change	It does not directly cut the main signal's bass. Does the key contain low frequencies, and is compression active? Lower Mix also makes differences less audible.
HPF history does not follow every drag	History refreshes after release; wait for HPF UPDATING to clear. EXT Key Gain updates live.
I hear the sidechain source	Check whether SC LISTEN is on; turn it off for normal output.
One side only / narrowed image	Return MONITOR to ALL, then check LR / MS settings.
Match is unavailable	Play representative non-silent material to establish a usable loudness reading.
Host briefly realigns after a Lookahead change	Latency changes, so the host may realign compensation. Also consider latency while monitoring a live recording.
Nothing shown below -90 dB	This is the Display range, not a cutoff of the audio.

For installation, updates and plug-in scanning, see the supplied Windows / macOS installation guide.

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github.com/Ziqing-Gu/qq-super-compression